

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from skimage.feature import hog
from skimage import exposure
import os
```

```
filename1 = '/content/images (1).jpg'
filename2 = '/content/images.jpg'
```

```
for f in [filename1, filename2]:
    if not os.path.exists(f):
        raise FileNotFoundError(f"'{f}' not found in {os.getcwd()}. Files here: {os.li
```

```
img1 = cv2.imread(filename1)
img2 = cv2.imread(filename2)
```

```
gray1 = cv2.cvtColor(img1, cv2.COLOR_BGR2GRAY)
gray2 = cv2.cvtColor(img2, cv2.COLOR_BGR2GRAY)
```

```
sift = cv2.SIFT_create()
keypoints1, descriptors1 = sift.detectAndCompute(gray1, None)
keypoints2, descriptors2 = sift.detectAndCompute(gray2, None)
```

```
sift_img1 = cv2.drawKeypoints(img1, keypoints1, None, flags=cv2.DRAW_MATCHES_FLAGS_DRAW_
sift_img2 = cv2.drawKeypoints(img2, keypoints2, None, flags=cv2.DRAW_MATCHES_FLAGS_DRAW_
```

```
fd1, hog_image1 = hog(
    gray1, orientations=9, pixels_per_cell=(8, 8),
    cells_per_block=(2, 2), visualize=True, block_norm='L2-Hys'
)
```

```
hog_image1_rescaled = exposure.rescale_intensity(hog_image1, in_range=(0, 10))
```

```
plt.figure(figsize=(15, 10))
```

```
<Figure size 1500x1000 with 0 Axes>
<Figure size 1500x1000 with 0 Axes>
```

```
plt.subplot(2, 3, 1)
plt.imshow(cv2.cvtColor(img1, cv2.COLOR_BGR2RGB))
plt.title('Image 1 - Original')
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(274.5), np.float64(182.5), np.float64(-0.5))
```

Image 1 - Original



```
plt.subplot(2, 3, 2)
plt.imshow(cv2.cvtColor(sift_img1, cv2.COLOR_BGR2RGB))
plt.title(f'Image 1 - SIFT ({len(keypoints1)} kp)')
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(274.5), np.float64(182.5), np.float64(-0.5))
```

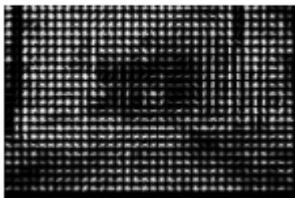
Image 1 - SIFT (352 kp)



```
plt.subplot(2, 3, 3)
plt.imshow(hog_image1_rescaled, cmap='gray')
plt.title('Image 1 - HOG')
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(274.5), np.float64(182.5), np.float64(-0.5))
```

Image 1 - HOG



```
plt.subplot(2, 3, 4)
plt.imshow(cv2.cvtColor(img2, cv2.COLOR_BGR2RGB))
plt.title('Image 2 - Original')
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(519.5), np.float64(259.5), np.float64(-0.5))
```

Image 2 - Original



```
plt.subplot(2, 3, 5)
plt.imshow(cv2.cvtColor(sift_img2, cv2.COLOR_BGR2RGB))
plt.title(f'Image 2 - SIFT ({len(keypoints2)} kp)')
plt.axis('off')
```

```
(np.float64(-0.5), np.float64(519.5), np.float64(259.5), np.float64(-0.5))
```

Image 2 - SIFT (1184 kp)



```
plt.tight_layout()
plt.savefig('sift_hog_output.png')
print("Saved sift_hog_output.png")
```

Saved sift_hog_output.png
<Figure size 640x480 with 0 Axes>

```
bf = cv2.BFMatcher()
matches = bf.knnMatch(descriptors1, descriptors2, k=2)
```

```
good_matches = []
for m, n in matches:
    if m.distance < 0.75 * n.distance:
        good_matches.append(m)
```

```
match_img = cv2.drawMatches(
    img1, keypoints1, img2, keypoints2, good_matches, None,
    flags=cv2.DrawMatchesFlags_NOT_DRAW_SINGLE_POINTS
)
```

```
match_img_rgb = cv2.cvtColor(match_img, cv2.COLOR_BGR2RGB)
```

```
plt.figure(figsize=(15, 6))
plt.imshow(match_img_rgb)
plt.title(f'SIFT Feature Matching ({len(good_matches)} good matches)')
plt.axis('off')
plt.tight_layout()
plt.savefig('sift_matching_output.png')
print("Saved sift_matching_output.png")
```

Saved sift_matching_output.png

SIFT Feature Matching (0 good matches)



```
print(f"Image 1: {len(keypoints1)} SIFT keypoints")
print(f"Image 2: {len(keypoints2)} SIFT keypoints")
print(f"Good matches found: {len(good_matches)}")
```

```
Image 1: 352 SIFT keypoints
Image 2: 1184 SIFT keypoints
Good matches found: 0
```